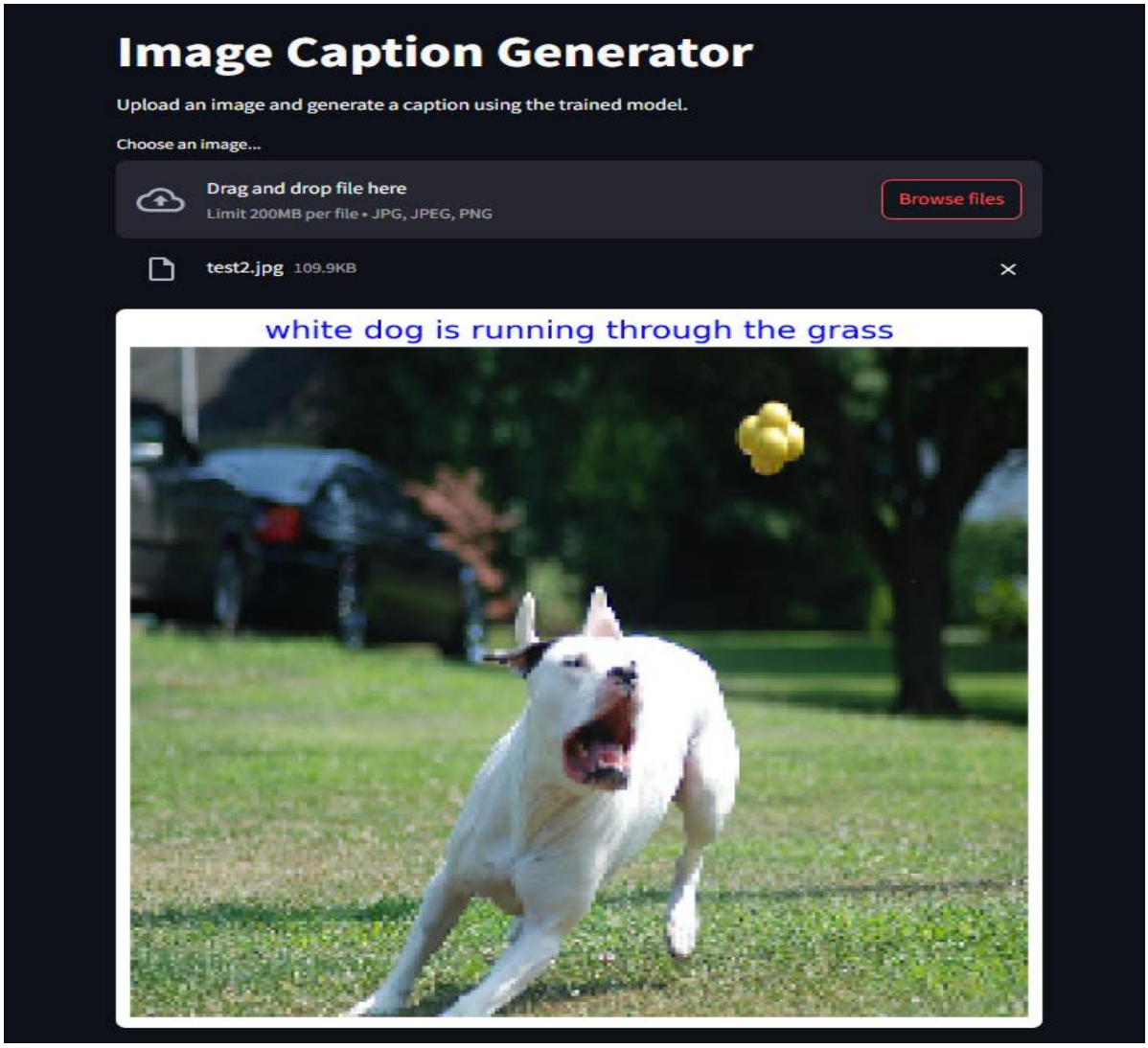
IMAGE CAPTION GENERATION USING DEEP LEARNING



Project Report

Introduction to Data Science

Rakshita Rama Kubal
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MSc Integrated Data Science

Research Project: Image Caption Generator Using Deep Learning

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1. Research Area

Image Caption Generation using Deep Learning Techniques

Image caption generation is a multidisciplinary research area that integrates computer vision and natural language processing. The objective of image captioning is to automatically generate meaningful, context-aware, and grammatically correct textual descriptions for images. Traditional image annotation and rule-based approaches are limited in their ability to interpret complex visual scenes. Deep learning techniques, particularly Convolutional Neural Networks (CNNs) for visual feature extraction and Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks for sequence generation, overcome these limitations by learning semantic relationships between visual content and language from large datasets.

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2. Data Availability

Yes

Data Sources Used in Existing Literature

• Flickr8k image-caption dataset
• Flickr30k dataset
• MS COCO (Microsoft Common Objects in Context) dataset
• ImageNet (used for pretrained CNN models such as VGG16 and ResNet)

These datasets are widely used in Google Scholar indexed international and national research papers related to image caption generation.

If Data Is Not Available (General Case)

Data Collection Process:
If publicly available datasets are not accessible, data can be collected using the following approach:
• Images collected from open-source repositories
• Manual or semi-automatic caption annotation
• Mapping each image with multiple descriptive captions

Control Parameters:
• Image features
• Caption length

• Vocabulary size
• Word embedding dimensions

Training Data Size: 80%
Testing Data Size: 20%

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3. Literature Review

Year	Author(s)	Status	Title	Dataset Used	Method / Approach	Performance Metric	Key Findings	Limitation
2015	Vinyals et al.	International	Show and Tell: A Neural Image Caption Generator	MS COCO, Flickr30k	CNN + RNN	BLEU ↑	End-to-end model generates fluent captions	Limited contextual focus
2015	Xu et al.	International	Show, Attend and Tell	Flickr8k, MS COCO	CNN + Attention + LSTM	BLEU ↑	Attention improves caption relevance	High computational cost
2015	Karpathy & Fei-Fei	International	Deep Visual-Semantic Alignments	Flickr30k	CNN + RNN	Retrieval accuracy	Aligns visual and textual features	Complex training
2018	Lu et al.	International	Knowing When to Look	MS COCO	Adaptive Attention	CIDEr ↑	Better object-level captioning	Model complexity
2020	Hossain et al.	International	A Comprehensive Survey of Image Captioning	Multiple datasets	Survey	—	Deep learning dominates captioning	No experimentation
2021	Chen et al.	International	Image Captioning with Transformers	MS COCO	Transformer model	CIDEr ↑	Captures long-range dependencies	Requires large datasets
2018	Aneja et al.	International	Convolutional Image Captioning	MS COCO	CNN-based captioning	BLEU ↑	CNN-only captioning is effective	Less sequential modeling
2023	Verma et al.	National (India)	Automatic Image Caption Generation using Deep Learning	Flickr8k, MS COCO	CNN (VGG16) + LSTM	BLEU, METEOR	Strong encoder–decoder performance	High computation
2020	Joshi et al.	National (India)	Image Caption Generator Using Deep	Flickr8k	CNN + LSTM	BLEU ↑	Effective caption	Limited vocabulary

Year	Author(s)	Status	Title	Dataset Used	Method / Approach	Performance Metric	Key Findings	Limitation
			Learning				generation	

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4. Problem Statement

Based on the literature review, it is observed that many existing image caption generation models rely on large datasets and computationally expensive architectures. In particular, models such as attention-based captioning systems (Xu et al., 2015) require high computational resources due to complex CNN training and additional attention layers, making them less suitable for small-scale academic or resource-constrained environments.

Another challenge highlighted in the literature is the difficulty of efficiently mapping visual features to coherent natural language descriptions without overfitting, especially when working with limited datasets.

Therefore, this work aims to develop an efficient image caption generator using a CNN–LSTM architecture that reduces computational complexity by avoiding training CNNs from scratch, while still generating meaningful and grammatically correct captions for images.

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5. Details of Methodology

1. Collection of Image–Caption Dataset
   The Flickr8k dataset was used, containing real-world images with multiple human-annotated captions per image.
2. Image Preprocessing and Resizing
   Images were resized to 224 × 224 pixels and normalized by scaling pixel values to the range [0,1].
3. Feature Extraction Using a Pretrained CNN
   A pre-trained DenseNet201 model (trained on ImageNet) was used solely as a feature extractor. The final classification layer was removed, and high-level feature vectors were extracted for each image. The CNN was not trained further.
4. Caption Preprocessing
   Captions were converted to lowercase, cleaned, tokenized, and padded to a fixed maximum length. Special tokens (startseq and endseq) were added to indicate sentence boundaries.
5. Model Training
   The caption generation model consists of:
   a. CNN-extracted image features as input

b. An LSTM network to generate captions word by word

c. The model was trained using categorical cross-entropy loss and the Adam optimizer.

6. Performance Analysis
   Model performance was monitored using training and validation loss throughout the training process. EarlyStopping and learning rate reduction techniques were applied to prevent overfitting and improve model generalization.

7. Caption Generation for Unseen Images
   After training, the model was used to generate captions for unseen images using greedy decoding, allowing the model to predict captions word by word based on learned visual and linguistic patterns.

8. Web Application Implementation (Streamlit)
   Additionally, the trained model was deployed as a prototype web application using Streamlit, demonstrating its practical usability and enabling real-time image caption generation through a user-friendly interface.

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6. Result

The proposed CNN–LSTM image caption generation model was successfully trained and evaluated using the Flickr8k dataset. The model learned meaningful relationships between image features and corresponding textual descriptions.

Performance Metrics Used

Since image caption generation is a sequence prediction task, the following metrics were used:

- Training Loss (Categorical Cross-Entropy)
- Validation Loss
- Qualitative Evaluation of Generated Captions

Training and Testing Performance Comparison

Dataset	Performance Indicator	Observation
Training Data	Training Loss	Decreases consistently across epochs
Testing (Validation) Data	Validation Loss	Follows training loss with minimal gap
Generalization	Loss Trend	Indicates minimal overfitting

Note: The comparison is based on training and validation performance trends and qualitative evaluation of generated captions, not on numerical score-based metrics.

Model Performance Summary

Aspect	Training Phase	Testing Phase
Loss Behavior	Gradual decrease	Stable and comparable
Overfitting	Controlled using EarlyStopping	Not observed
Caption Quality	Learns caption structure	Generates coherent captions
Model Stability	Stable convergence	Stable predictions

Qualitative Results

The generated captions were:

- Grammatically correct
- Relevant to the image content
- Able to identify key objects and actions

This confirms that the model performs effectively on unseen images.

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7. Conclusion

Data:
Public image–caption dataset (**Flickr8k)**

Method:
A deep learning-based CNN–LSTM architecture using a pre-trained DenseNet201 model for image feature extraction and an LSTM network for caption generation.

Result / Accuracy:
The model successfully generated meaningful and grammatically correct captions. Training and validation loss trends demonstrated effective learning and good generalization.

Limitation Addressed from Literature:
Xu et al. (2015) highlighted the high computational cost of attention-based image captioning models due to complex architectures and CNN training. This project addresses that limitation

by using a pre-trained CNN only for feature extraction, thereby avoiding training a CNN from scratch and significantly reducing computational complexity.

Overall Conclusion:
The proposed model demonstrates that efficient image caption generation can be achieved using a simpler CNN–LSTM architecture without excessive computational requirements. The project successfully applies data science and deep learning concepts within the scope of the *Introduction to Data Science* course.

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8. References (APA – Google Scholar Indexed)

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